ManoVarta: Multilingual Conversational Screening for Mental Health

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Abstract

ManoVarta is a multilingual conversational screening system that turns open-ended English, Hindi, and Hinglish dialogue into PHQ-9 and GAD-7 item scores, evidence summaries, and safety decisions. The core lesson of the project is that fluent prompting alone is not enough: early direct-scoring baselines sounded plausible but covered almost none of the questionnaire in a usable way. The final system therefore uses an evidence-first architecture with application-controlled dialogue steering, item-level coverage tracking, a separate safety stack, and a specialized extractor path. On the archived live runtime benchmark, ManoVarta reaches 0.783 coverage completeness, 0.639 exact match, 0.251 macro-F1, 0 parse failures, and 1.0 precision / 1.0 recall for safety. The deployed runtime also supports public web access, browser and cloud voice I/O, and a budget-safe split between a CPU conversational service and a remote Aya extractor.

1 Introduction

Mental-health screening instruments such as PHQ-9 and GAD-7 are valuable because they are interpretable, clinically familiar, and easy to score (Kroenke et al., 2001; Spitzer et al., 2006). Their weakness is interaction quality: a rigid questionnaire often feels evaluative rather than conversational. ManoVarta targets that gap by treating screening as guided dialogue while still requiring item-level outputs, explicit uncertainty, and a separate safety decision.

The project did not improve by simply switching to larger models. Our early heuristic and direct-prompt baselines revealed the same failure mode: the system could produce natural text while leaving most questionnaire items untouched. Progress came from adding structure. The final design separates (i) dialogue control, (ii) evidence extraction, (iii) questionnaire scoring, and (iv) safety escalation. This report focuses on that measured path from weak baseline to deployable multilingual runtime.

The project makes three contributions. First, it builds a multilingual screening runtime grounded in PHQ-9 and GAD-7, with English and Hindi as required languages and Hinglish treated as a real robustness condition. Second, it shows that offline extraction strength and deployed safety are not the same objective. Third, it documents a practical deployment split in which the application owns steering, the live language layer stays inexpensive, and the heavier extractor remains available behind a remote or asynchronous interface.

2 Task and Data

The target output is a structured screening summary: PHQ-9 and GAD-7 item scores in the 0–3 range, supporting evidence, unresolved items, and a safety label (`none`, `review`, or `urgent`). We treat the system as a screening aid rather than a diagnostic or therapeutic agent (Abd-Alrazaq et al., 2020; Dosovitsky et al., 2021). Safety is evaluated separately because crisis-detection failure is qualitatively different from ordinary questionnaire error (Zirikly et al., 2019; Pichowicz et al., 2025).

The data pipeline has two layers. The first is a bilingual gold compliance pack of 60 audio-backed sessions (30 English and 30 Hindi). Each session includes transcript, metadata, and three label files: `annotator_a`, `annotator_b`, and `adjudicated`. The adjudication summary over the full pack reports 0.927 exact agreement and Cohen's $\kappa = 0.7319$. The second layer consists of model-development exports used for extractor, follow-up, and safety training. Hinglish is used for robustness evaluation and seed/runtime testing rather than as a gold split. This distinction matters because multilingual voice support was a core product goal, but the strongest source-matched benchmark remained English.

We report four metric families. **Discourse effectiveness** uses coverage completeness, exact match, macro-F1, and parse failures. **Safety accuracy** uses precision and recall. **Disclosure efficiency** measures the user turns needed before an item score stabilizes. **Runtime viability** uses latency and successful endpoint deployment. These metrics reflect the real project requirement: the system must converse, score, and stay safe rather than merely maximize one offline classifier number.

3 System

The final system is an evidence-first conversational pipeline. The application-side dialogue planner tracks topic state, coverage debt, reopen/summary conditions, and safety-related constraints. It decides which topic or item needs attention next and then asks the live language layer to phrase the next turn naturally. This makes the application, not the model, the true controller.

The evidence path is separate from text generation. Candidate symptom evidence is extracted from the conversation, mapped to PHQ-9 and GAD-7 items, and only then converted into item-level scores. This design avoided the common failure case where a model produced a clinically styled summary without enough grounded support. Aya became the extractor backbone because it was the first model family in the project to produce a strong archived multilingual result rather than a narrow probe success (Cohere For AI, 2024). Later deployment work kept that extractor logic but moved the public runtime to a cheaper split: the web service handles live conversation on CPU, while the heavier trained Aya path is reached remotely or asynchronously when richer extraction is needed.

Safety is also handled outside the main generator. ManoVarta combines rules, a promoted local safety checkpoint, and corroboration logic. This hybrid design proved much more reliable than trusting a single conversational model to both interview and assess risk. For deployment, the public runtime uses Vertex/Gemini for live multilingual phrasing, a remote Aya extractor for stronger evidence extraction, and a local safety checkpoint for low-latency risk filtering. The result is not a monolithic model demo but a controller-oriented screening product.

4 Experiments and Results

Table 1 summarizes the most important project milestones. The heuristic runtime achieved a superficially decent exact-match score, but only 0.022 coverage, which made it unusable in practice. The offline Aya extractor was the first system that behaved like a true screening engine, pushing coverage above 0.9 with zero parse failures. However, that extractor-only setup still failed end-to-end safety. The deployed objective therefore shifted from “best extractor” to “best safe screening runtime.”

The final live runtime reaches 0.783 coverage completeness, 0.639 exact match, 0.251 macro-F1, and 0 parse failures, while preserving 1.0 precision and 1.0 recall on the held-out runtime safety evaluation. Language-wise, the deployed runtime reaches 0.844 coverage in English, 0.719 in Hindi, and 0.786 in Hinglish. English became the strongest slice only after later controller fixes and verifier-style repair for ambiguity in anxiety and functioning items; Hindi remains the weakest split because sparse symptom mentions are still easier to miss than in English or code-mixed Hinglish.

An important project observation is that exact match alone was misleading early on. The heuristic baseline looked acceptable by exact match because it abstained on most items. Coverage and parse stability exposed the real weakness. The opposite problem appeared in the strong offline extractor: high item coverage without a safe deployed interaction policy. The final system therefore optimizes for a balanced target: enough coverage to produce useful screening outputs, enough control to prevent looping or premature closure, and enough safety calibration to justify live use as a screening support tool.

The latest controller refinements also improved practical multilingual chat quality. The runtime now suppresses stale anxiety carryover from neutral or physical openings, avoids repeated Hindi sleep/mood probes, and keeps Hinglish low-energy narratives from jumping into anxiety too early. These fixes matter because conversation quality directly affects whether the system can gather the evidence needed to score PHQ-9 and GAD-7 reliably.

System milestone	Coverage	Exact	Macro-F1	Safety P/R	Main role
Heuristic seed runtime	0.022	0.635	0.058	1.000 / 0.604	rules + lexicons
Saved local Qwen probe (n=3)	0.429	0.667	0.129	0.000 / 0.000	early compact-model probe
Aya offline extractor	0.913	0.562	0.272	0.000 / 0.000	multilingual evidence extractor
Aya continuation rerun	0.943	0.614	0.297	1.000 / 1.000	strongest archived held-out run
Hybrid runtime validation	0.538	0.712	0.295	1.000 / 1.000	safety-first runtime checkpoint
Final live runtime	**0.783**	**0.639**	**0.251**	**1.000 / 1.000**	**deployed screening runtime**

Table 1: Main milestone comparison. The strongest offline extractor and the strongest deployed system are not the same point on the curve. The final runtime gives up some raw coverage relative to the best Aya extractor in exchange for safer and more robust end-to-end behavior.

5 Deployment and Discussion

The final deployment is not just an offline benchmark. ManoVarta is available as a public Cloud Run service with browser chat, voice endpoints, live configuration inspection, and a remote Aya extractor service. The public runtime currently uses a budget-safe split: CPU hosting for the conversational layer, a remote extraction path for trained Aya, and a local safety checkpoint. This architecture emerged from cost and reliability constraints. Keeping a large dual-model stack online continuously was not practical, so the project moved the heavy extractor behind a separate boundary while preserving the controller logic and multilingual screening behavior.

This deployment story reinforces the main technical lesson: conversational screening should not be treated as a single-prompt generation task. The best-performing product version came from explicit control over coverage, topic switching, safety escalation, and summary timing. In practice, the application had to know when to continue naturally, when to gather missing evidence, and when not to let the language model drift into repetitive anxiety loops.

6 Limitations and Conclusion

Several limitations remain. The bilingual gold pack is careful and useful, but still modest; these results are system-validation results, not clinical-validation claims. Hindi remains weaker than English because the underlying source material is less source-matched to mental-health interviewing. Hinglish is robust enough to be useful, but severity calibration on some mixed cues still lags behind pure English. Finally, perfect held-out safety precision and recall in this project do not guarantee real-world safety without human oversight.

Within the assignment scope, however, ManoVarta demonstrates a clear result: multilingual mental-health screening becomes much more reliable when the system is structured around evidence, controller logic, and separate safety handling instead of raw conversational fluency. The final live runtime is not the numerically strongest extractor built during the project, but it is the strongest deployable screening system produced by the project: multilingual, voice-capable, safety-consistent, parse-stable, and grounded in PHQ-9 and GAD-7.

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